miniRT

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November 15, 2025

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1 Linear algebra

1.1 Vector

A vector is a mathematical object described by it's components, it has 1 component for each dimension, so a 3D vector has 3 components.
It has the following operations for 2 vectors, $\vec{v}$, $\vec{w} \in \mathbb{R}^3$ and a scalar $\lambda \in \mathbb{R}$.

Addition	$\vec{v} + \vec{w}$	$=$	$(v_x + w_x, v_y + w_y, v_z + w_z)$
Substraction	$\vec{v} - \vec{w}$	$=$	$(v_x - w_x, v_y - w_y, v_z - w_z)$
Scale	$\lambda\vec{v}$	$=$	$(v_x \cdot \lambda, v_y \cdot \lambda, v_z \cdot \lambda)$
Dot product	$\vec{v} \cdot \vec{w}$	$=$	$v_x w_x + v_y w_y + v_z w_z$
Inner-product	$\langle \vec{v}, \vec{w} \rangle$	$=$	$\vec{v} \cdot \vec{w}$
Cross product	$\vec{v} \times \vec{w}$	$=$	$(v_y w_z - v_z w_y, v_z w_x - v_x w_z, v_x w_y - v_y w_x)$
Magnitude	$\|\|\vec{v}\|\|$	$=$	$\sqrt{v_x^2 + v_y^2 + v_z^2}$
Normalization	$\hat{v}$	$=$	$\vec{v} \,/\, \|\|\vec{v}\|\|$

1.2 Matrix

2 Vector space objects

2.1 Ray

For a point $\dot{O} \in \mathbb{R}^3$, the origin of the ray, a vector $\hat{N} \in \mathbb{R}^3$, the normalalized direction vector of the ray, and a scalar for the direction vector, $t \in \mathbb{R}$. We define the normal equation of a line as :

$$P(t) = \dot{O} + t\hat{N}$$

2.2 Plane

For a point $\dot{O} \in \mathbb{R}^3$, the origin of the plane, a vector $\hat{N} \in \mathbb{R}^3$, the normalized orthogonal vector of the plane and a arbitrary point $\dot{P} \in \mathbb{R}^3$. $\dot{P}$ is a point of the plane if :

$$N_x(P_x - O_x) + N_y(P_y - O_y) + N_z(P_z - O_z) = 0$$

2.3 Sphere

For a point $\dot{O} \in \mathbb{R}^3$ the origin of the sphere, a scalar $r \in \mathbb{R}$, the radius of the sphere and an arbitrary point $P \in \mathbb{R}^3$. $\dot{P}$ is a point of the sphere if :

$$\langle (\dot{P} - \dot{O}), (\dot{P} - \dot{O}) \rangle = r^2$$

2.4 Circle

For a point $\dot{O} \in \mathbb{R}^3$, the origin of the circle, a vector $\hat{N} \in \mathbb{R}^3$, the normalized orthogonal vector of the circle and a arbitrary point $\dot{P} \in \mathbb{R}^3$. $\dot{P}$ is a point of the circle if :

2.5 Cylinder

For a point $\dot{O} \in \mathbb{R}^3$ the origin of the sphere, a a vector $\hat{N} \in \mathbb{R}^3$ the normalized direction vector, two scalars r and $h \in \mathbb{R}$, respectively the radius and the height of the cylinder. We define the

normal equation of a cylinder as :

$$||(P - O) - \langle (P - O), \hat{N} \rangle \hat{N}||^2 = r^2$$

where

$$|\langle (P - O), \hat{N} \rangle| \leq \frac{h}{2}$$

3 Raytracer

3.1 Camera

Given informations :

- $\mathbf{C_{pos}}$: 3D Position as a point.
- $\mathbf{C_{dir}}$: 3D Direction as a normalized vector.
- fov° : Vertical field of view in degrees, ranging from 0° to 180°.

Deduced informations :

- $\mathbf{C_{fov}}$: Vertical field of view in radians, ranging from 0 to π radians We can define it with the following fomula :

$$\mathrm{C_{fov}} = \mathrm{fov}^\circ \cdot \frac{\pi}{180^\circ}$$

- $\mathbf{C_{flen}}$: Focal length that can be calculated as follows :

$$\mathrm{C_{flen}} = \tan\left(\frac{\mathrm{C_{fov}}}{2}\right)$$

3.2 Ray intersections

3.2.1 Plane

$$P(t) = \frac{(\dot{P}_o - \dot{R}_o) \cdot \hat{P}_d}{\hat{R}_d \cdot \hat{P}_d}$$

3.2.2 Circle

3.2.3 Sphere

3.2.4 Cylinder

3.2.5 Cone